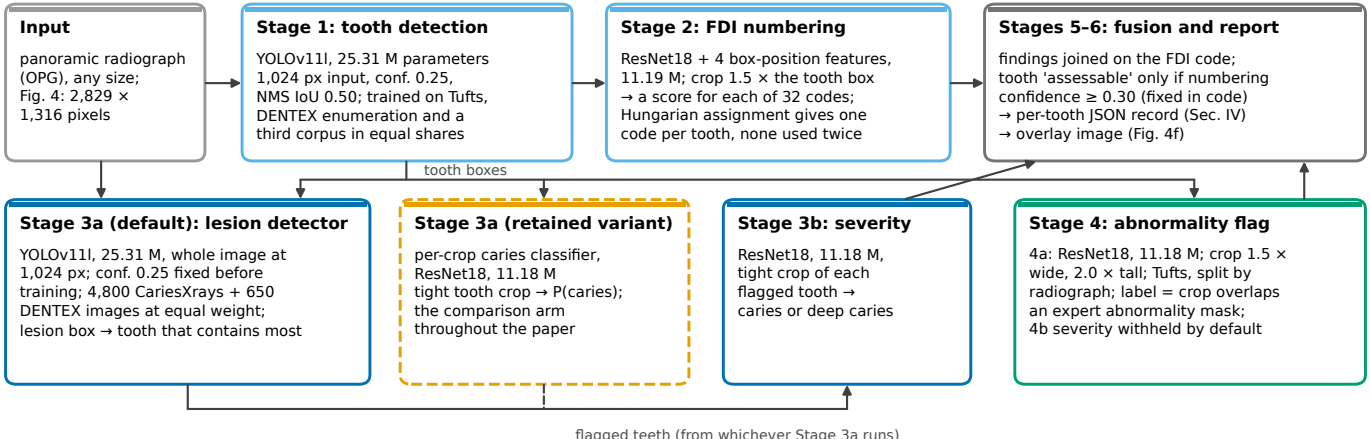
Input
panoramic radiograph (OPG), any size; Fig. 4: 2,829 × 1,316 pixels
Stage 1: tooth detection
YOLOv11l, 25.31 M parameters 1,024 px input, conf. 0.25, NMS IoU 0.50; trained on Tufts, DENTEX enumeration and a third corpus in equal shares
Stage 2: FDI numbering
ResNet18 + 4 box-position features, 11.19 M; crop 1.5 × the tooth box → a score for each of 32 codes; Hungarian assignment gives one code per tooth, none used twice
Stages 5–6: fusion and report
findings joined on the FDI code; tooth 'assessable' only if numbering confidence ≥ 0.30 (fixed in code) → per-tooth JSON record (Sec. IV) → overlay image (Fig. 4f)
tooth boxes
Stage 3a (default): lesion detector
YOLOv11l, 25.31 M, whole image at 1,024 px; conf. 0.25 fixed before training; 4,800 CariesXrays + 650 DENTEX images at equal weight; lesion box → tooth that contains most
Stage 3a (retained variant)
per-crop caries classifier, ResNet18, 11.18 M tight tooth crop → P(caries); the comparison arm throughout the paper
Stage 3b: severity
ResNet18, 11.18 M, tight crop of each flagged tooth → caries or deep caries
Stage 4: abnormality flag
4a: ResNet18, 11.18 M; crop 1.5 × wide, 2.0 × tall; Tufts, split by radiograph; label = crop overlaps an expert abnormality mask; 4b severity withheld by default
flagged teeth (from whichever Stage 3a runs)